



Promotional effect of Mn-doping on the structure and performance of spinel ferrite microspheres for CO hydrogenation

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ABSTRACT

A series of uniform microspheres of spinel-type Mn-doped ferrites ($\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$) with different Fe/Mn ratios were employed for the Fischer-Tropsch-to-olefins reaction (FTO). The highest selectivity of 54.4% and the space-time-yield (STY) of $73.0\text{ g}\cdot\text{kg}_{\text{cat}}^{-1}\cdot\text{h}^{-1}$ to lower olefins ($\text{C}_2\text{-C}_4$) were achieved over an Fe-Mn catalyst with 13 wt% Mn. A panoramic view of structure evolution of $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ was revealed using multiple *in/ex situ* techniques. We demonstrated that the expansion of lattice spacing induced by Mn exerted remarkable effects on the stabilization of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ and $\epsilon\text{-Fe}_2\text{C}$ during activation and reaction. Moreover, Mn endows Fe catalysts with strong surface basicity and consequently enhances the adsorption and dissociation of CO; while weakening the H availability and readsorption of olefinic intermediates. In addition, the impeded carburization of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ and the reduction in particle size of catalysts jointly contribute to the volcano curve in the catalytic activity. The insights presented could guide the rational design of high performance catalysts via heteroatom doping.

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1. Introduction

Lower olefins ($\text{C}_2\text{-C}_4$), currently produced primarily from steam cracking and fluid catalytic cracking, are essential cornerstones for chemical industry [1–4]. From both environmental and economic perspectives, nowadays, intensive research interests in alternative technologies and high-performance catalysts for olefin synthesis have been aroused [3,5–11]. Syngas (CO and H_2) conversion is a pivotal platform reaction which gives impetus to the clean and efficient utilization of non-oil fuels (e.g., natural gas, coal, biomass) [12,13]. With the rapid depletion of oil resources, the direct conversion of syngas into lower olefins via Fischer-Tropsch-to-olefins (FTO) becomes a feasible route [1,5,14]. However, the Anderson-Schulz-Flory (ASF) distribution for Fischer-Tropsch (FT) products usually imposes a limitation on the lower-olefin selectivity [3]. The objective of FTO research is to maximize lower olefins selectivity while reducing the formation of methane and excess CO_2 [1].

Extensive efforts in the development of modified FT catalysts have been devoted to the enhancement of the lower olefin selectivity [2,3,5–7,15]. Among all the active metals, iron is easily accessible, inexpensive, poison-tolerant and less reactive to secondary hydrogenation of olefins [16,17]. Besides, owing to high water gas shift (WGS) activity, Fe-based catalysts are particularly suitable for the conversion of CO-rich syngas derived from coal or biomass. Nevertheless, low efficiency, poor chemical and mechanical stability under harsh reaction conditions (high temperature, pressure) hinder its industrialization [3].

In the past several decades, different phases of Fe (i.e., Fe_3O_4 , metallic Fe, Fe carbides) produced during reaction have been proposed to be the active phase for FTS [18,19]. The formation of iron carbides (FeC_x), primarily consisting of carbon atoms occupying interstitial sites of the iron lattice, is widely recognized to be necessary for the FTS reaction [20,21]. Stable (Pseudo-) cementite ($\theta\text{-Fe}_3\text{C}$), Hägg carbide ($\chi\text{-Fe}_5\text{C}_2$), Eckström-Adcock Fe_7C_3 and metastable $\epsilon\text{-Fe}_2\text{C}/\epsilon'\text{-Fe}_{2.2}\text{C}$ are well-known carbides observed in FTS [2,20,22]. $\chi\text{-Fe}_5\text{C}_2$ has proven to be the most active and selective in FTO, but susceptible to oxidation by water. The carbon-lean $\theta\text{-Fe}_3\text{C}$ phase, whose formation is always associated with a large amount of carbonaceous species, is considered as the deactivated

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phase [23,24]. Based on theoretical predictions, ϵ -Fe₂C/ ϵ' -Fe_{2.2}C is thermodynamically more stable and active in the reaction at low temperatures, while it showed higher oxidation tolerance than χ -Fe₅C₂ in FTS [20,25]. The role of the specific carbide species remains controversial to date. In addition, the metallic Fe and magnetite, usually analyzed in spent Fe-based catalysts, still cannot be excluded as the possible active phases [19,21].

Generally, the electronic and geometric structure could be modified with the addition of promoters, leading to the variation in properties such as surface acidity and basicity, diffusion, adsorption and desorption of reactants/products, reduction and phase transformation properties, deactivation behavior, etc. Several studies have demonstrated that the addition of various promoters such as K, Na, S, and Mn to Fe-based FTS catalysts is effective to improve the olefin synthesis [6,26–28]. Especially, Mn, as an efficient promoter, has been widely used in FTO catalysts on the basis of following advantages: (i) resistance to deactivation; (ii) enhancement of the selectivity to lower olefins; (iii) improvement of activity for Mn-doped catalysts; (iv) facile preparation of spinel-like Mn_xFe_{3-x}O₄ with fine-tuned electrical and magnetic properties.

However, the effect of Mn on the performance of Fe catalysts is still disputable. Lohitharn reported that the addition of Mn could significantly improve the activity and stability of Fe-based catalysts, enhance the lower olefins selectivity and suppress the formation of methane [29]. Moreover, a certain amount of Mn was observed to improve the dispersion of iron oxide and thus promote the intrinsic activity [30]. Besides, Jensen found that the doping of Mn can increase the surface alkalinity of the catalyst and facilitate the phase transformation of iron oxide to iron carbides [31]. Recently, it was reported that the Mn promoter affected the electronic state of surface carbonaceous species, leading to the formation of θ -Fe₃C whose content is positively correlated with the selectivity [32]. However, Li *et al.* [33] reported that the introduction of Mn atoms did not significantly change the lower olefins selectivity but increase the production of long chain hydrocarbons. However, the interaction mechanism between Mn and Fe oxides and the structure-performance relationship of Fe-Mn catalysts are hitherto unclear and attracting ongoing concern in academia.

Barraut *et al.* [34] reported that the beneficial effect of Mn could be largely related to the preparation method, and more specifically to the formation of Fe_xMn_yO_z precursor compounds. The synthesis of monodispersed nanoparticles with controlled size and shape has been intensively pursued for their technological and fundamental scientific importance. However, traditional preparation methods of Fe-Mn binary oxides are hard to control the size and shape. Spinel ferrites (MFe₂O₄, M = Fe, Mn, Zn, or Co) are among the most concerned magnetic materials and have been widely used in electromagnetic devices, information storage, magnetic resonance imaging (MRI), drug-delivery technology, and biosensors, etc [35–37]. The solvothermal reduction method has previously been used to successfully fabricate a series of uniform, monodisperse, and single-crystalline ferrite MFe₂O₄ (M = Fe, Mn, Zn, or Co) microspheres [28,35].

In this paper, a one-step solvothermal approach was employed to synthesize a series of spinel-like Mn_xFe_{3-x}O₄ microspheres with different Fe/Mn ratios. By adjusting the relative content of Fe and Mn, uniform Fe-Mn microspheres were prepared by modified reduction reaction between precursors (FeCl₃, MnCl₂) and ethylene glycol, and the FTO performance was evaluated over as-prepared spinel-like Fe-Mn microspheres. The structure evolution of Fe-Mn microspheres catalysts during activation and reaction was systematically investigated by multiple techniques such as scanning electron microscopy-energy dispersive X-ray spectroscopy (SEM-EDS), H₂-temperature-programmed reduction (H₂-TPR), Raman spectroscopy, X-ray photoelectron spectroscopy (XPS), *ex/in situ* X-ray diffraction (XRD) and Mössbauer spectroscopy. Finally, we eluci-

dated the intrinsic relationship between Mn dopant and the structure of Fe-Mn microspheres. This work contributes to an in-depth understanding of the complicated structure-performance relationship of bimetallic Fe-based catalysts for the FTO reaction.

2. Experimental section

2.1. Catalyst preparation

The spinel ferrite microspheres were prepared through a solvothermal method in a stainless-steel hydrothermal autoclave with a Teflon chamber [28,35,38]. The nominal Mn content in metal ions (0%, 5%, 33%, 80%) was adjusted by varying the relative ratio of Fe/Mn with a constant ion concentration. Briefly, FeCl₃·6H₂O (AR, Sinopharm) and MnCl₂·4H₂O (AR, Sinopharm) were dissolved in ethylene glycol with a total ion concentration of 0.1 M. After stirring for 30 min, sodium acetate was added for electrostatic stabilization to prevent particles from agglomeration. Next, the mixture was stirred vigorously for 1 h followed by its sealing in a Teflon lined stainless-steel autoclave (100 mL). The autoclave was heated to 473 K and maintained for 8 h, and then cooled down to room temperature naturally. Subsequently, the as-prepared precursors were washed successively with ethanol and water for several times, respectively, and then collected by a magnet, dried and stored in vacuum at room temperature. The samples were denoted as Fe-xMn, and x% is the actual atomic percentage of Mn in total metal ions (x = 0, 2, 13, 31). Manganous oxide (MnO), prepared without the addition of Fe precursor by the same solvothermal procedures, was employed as a reference for temperature-programmed reduction in H₂ (H₂-TPR).

2.2. Characterization

2.2.1. N₂ physisorption

The Brunauer-Emmett-Teller (BET) surface areas, pore volumes and diameters of as-prepared catalysts were measured with N₂ adsorption isotherms at 77 K by an ASAP-2010 system from Micromeritics after the samples were outgassed at 463 K and 1 mmHg for 6 h.

2.2.2. Inductively coupled plasma mass spectrometry (ICP-MS)

The actual elemental content of Fe and Mn was determined by inductively coupled plasma mass spectrometry (Perkin Elmer, NexION 300).

2.2.3. Scanning electron microscopy-energy dispersive X-ray spectroscopy (SEM-EDS)

The morphologies and elemental distributions of the catalysts were characterized using a FEI high-resolution FESEM (NOVA Nano SEM450) under the scanning electron (SE) mode equipped with an energy dispersive X-ray spectrometer. The accelerating voltage was set at 3 kV with a working distance of 5 mm. EDS mapping and the single-line profiles were obtained in a ZEISS GeminiSEM 500 equipped with an X-Max Extreme 100 mm² SDD windowless detector (Oxford Instruments).

2.2.4. X-ray diffraction (XRD)

X-ray powder diffraction (XRD) patterns were recorded by a Bruker D8 advance diffractometer using Cu-K α radiation (λ = 1.5406 Å) with an accelerated voltage of 40 kV and detector current of 40 mA. The XRD patterns of fresh catalysts were obtained with a step of 0.02° and a scanning rate of 0.24°/min, while those of spent catalysts were collected with the same step and scanning rate of 3°/min in a range of 10–80°. The grain size of the microspheres was calculated from the full-width at

half-maximum of Fe_3O_4 (3 1 1) diffraction at $2\theta = 35.5^\circ$ by the Scherrer's equation Eq. (1).

$$D = \frac{0.89\lambda}{\Delta \cos \theta} \quad (1)$$

where D : crystal size, λ : wavelength of X-rays, Δ : FWHM of diffraction peak, and θ : angle corresponding to the peak.

The lattice parameters were calculated with the software TOPAS for Rietveld refinement. *In situ* XRD measurements were carried out using the same instrument with an Anton Paar XRK-900 reaction chamber. The sample was heated in 5% H_2/Ar (50 mL/min) from room temperature to 623 K with a ramping rate of 5 K/min and then maintained at 623 K for 5 h. During *in situ* XRD, diffraction patterns were recorded after reaching a prespecified temperature and held for 30 min unless otherwise specified, while the scanning rate was 10°/min.

2.2.5. Raman spectroscopy

The laser Raman spectroscopy (LabRAM HR, Horiba J.Y., France) was equipped with a high-grade Leica microscope (long working distance objective 50x) and a visible 514.5 nm Ar^+ laser. The sample was placed in an in-situ cell under the protection of Ar with a flow rate of 30 mL/min. The single crystal silicon (520.7 cm^{-1}) was used for position correction. The confocal hole size was set to 200 μm for all the tests. The attenuator with a decay factor of 0.6 was used for the tests to avoid the signal overflow and sample damage. The exposure time was set to 40 s for each spectrum. The scattered light was collected at 180° (backscattering plane) and collected by a deep-depleted thermoelectrically cooled CCD array detector (1024 $\times$ 256 pixels, 26 mm in size).

2.2.6. Temperature-programmed reduction in H_2 (H_2 -TPR)

Prior to the experiment, the sample (50 mg) was purged with Ar (30 mL·min⁻¹) at 423 K for 1 h to remove the physical adsorbed H_2O . H_2 -TPR was performed using a micro fixed-bed reactor connected with a thermal conducted detector (TCD). The temperature was ramped from 303 K to 398 K with a rate of 20 K/min and then ramped to 1098 K with a rate of 10 K/min in a 10% H_2/Ar flow of 50 mL/min. A rough quantitative analysis of H_2 consumption of all catalysts was performed according to the TPR profiles.

2.2.7. X-ray photoelectron spectroscopy (XPS)

X-ray photoelectron spectroscopy (XPS) spectra were collected with a PHI 5000 Versaprobe spectrometer using a monochromatic Al-K radiation source (1486.6 eV, a passing energy of 58.7 eV). The base pressure of the instrument was approximately 2.0×10^{-8} Pa. The binding energies (BEs) were calibrated referencing to adventitious C 1s at 284.8 eV. The spent catalysts were passivated in 1.0% O_2/N_2 for 5 min subsequently, and then transferred to XPS chamber quickly. During sputtering, Ar pressure was maintained at 8.0×10^{-7} Pa, and the Argon ion (Ar^+) sputter etching was operated with a beam energy of 3.0 kV for 2 min.

2.2.8. Mössbauer spectroscopy

Room-temperature ^{57}Fe Mössbauer spectra were recorded using a WissEl/MSPCA spectrometer. The γ -ray radioactive source was ^{57}Co in a Rh matrix at 298 K; the transmitted photons were detected using a proportional counter. The velocity scale of the Mössbauer spectra is given with respect to α -Fe at 298 K.

2.3. Catalytic performance

The FTO reaction was performed in a stainless-steel continuous fixed-bed micro-reactor with a length of 450 mm, inner and external diameters of 6 and 9 mm. The Fe-Mn microspheres were first reduced at 623 K under H_2 (30 mL/min) for 5 h. Prior to the reac-

tion, the reaction system was pressurized to 2.0 MPa with syngas ($\text{H}_2/\text{CO} = 1$). The reaction temperature was monitored by a K-type thermocouple inserted into the catalyst bed, and time-on-stream of 20 h was taken for each measurement if without emphasis. The catalytic performances were evaluated under the following conditions: 533 K, gas hourly space velocity (GHSV) = 4000 h⁻¹, ratio of $\text{H}_2/\text{CO} = 1$, and 25 mg catalyst diluted with 75 mg SiC. The blank test was implemented using SiC, and CO conversion of less than 0.45% was detected under the same conditions. The effluent was analyzed using an on-line gas chromatograph (Shanghai Ruimin GC2060) equipped with a packed column (5A molecular sieve), a capillary column (Agilent HP-Plot Q) and a packed column (TDX-01) connected to a thermal conductivity detector (TCD) and two flamed ionization detectors (FID), respectively. Product selectivity is calculated on a carbon basis.

2.4. Calculations

CO conversion (X_{CO}) was determined by the summation of carbon number in all products (Eq. (2)).

$$X_{\text{CO}} (\%) = \frac{Q_{\text{out}} \sum v_i C_i}{Q_{\text{in}} v_{\text{CO}}} \times 100\% \quad (2)$$

Here, Q_{in} is the total volume flow rate of syngas per unit of time, mL/min. Q_{out} is the volume flow rate of the outlet stream, mL/min. v_i is the volume fraction of component i in the outlet stream. C_i is the carbon number of component i . v_{CO} is the concentration of CO in the feed gas, equal to p_i/p_0 , p_i : partial pressure of reactants, p_0 : total pressure in the system.

The selectivity towards each product was calculated based on Eq. (3):

$$S_i (\%) = \frac{Q_{\text{out}} v_i C_i}{Q_{\text{in}} v_{\text{CO}} X_{\text{CO}}} \times 100\% \quad (3)$$

The iron time yield (FTY), which was calculated based on moles of CO converted to hydrocarbons (excluding CO_2) per weight of iron (g) per unit time (s), was expressed as Eq. (4):

$$\text{FTY} (\text{mol} \cdot \text{g}_{\text{Fe}}^{-1} \cdot \text{s}^{-1}) = \frac{Q_{\text{in}} v_{\text{CO}} X_{\text{CO,HCs}}}{V_m m_{\text{cat}} \sigma} \quad (4)$$

m_{cat} - mass of catalyst in the reactor bed, g.

σ - loading amount of Fe element in the catalyst.

$X_{\text{CO,HCs}}$ - CO conversion to hydrocarbons, excluding CO_2 .

V_m - the molar volume of an ideal gas at standard temperature and pressure, $V_m = 22400 \text{ mL/mol}$.

The space-time yield (STY) of C_2 - C_4 olefins was calculated through Eq. (5):

$$\text{STY}_{\text{C}_2-\text{C}_4} \left(\text{g} \cdot \text{kg}_{\text{cat}}^{-1} \cdot \text{h}^{-1} \right) = \frac{Q_{\text{out}} \sum_{i=2}^4 v_i M_i}{V_m m_{\text{cat}}} \quad (5)$$

M_i is the molecular weight of component i .

3. Results

3.1. Structure of the fresh Fe-Mn microspheres

To observe the morphology and structure of Fe-Mn microspheres, SEM, Raman, and XRD of all Fe-Mn microspheres were conducted. From the SEM images (Fig. 1), the microspheres are uniform, monodispersed, and composed of small nanocrystals. With increasing the Mn content from 0% to 31%, the microspheres

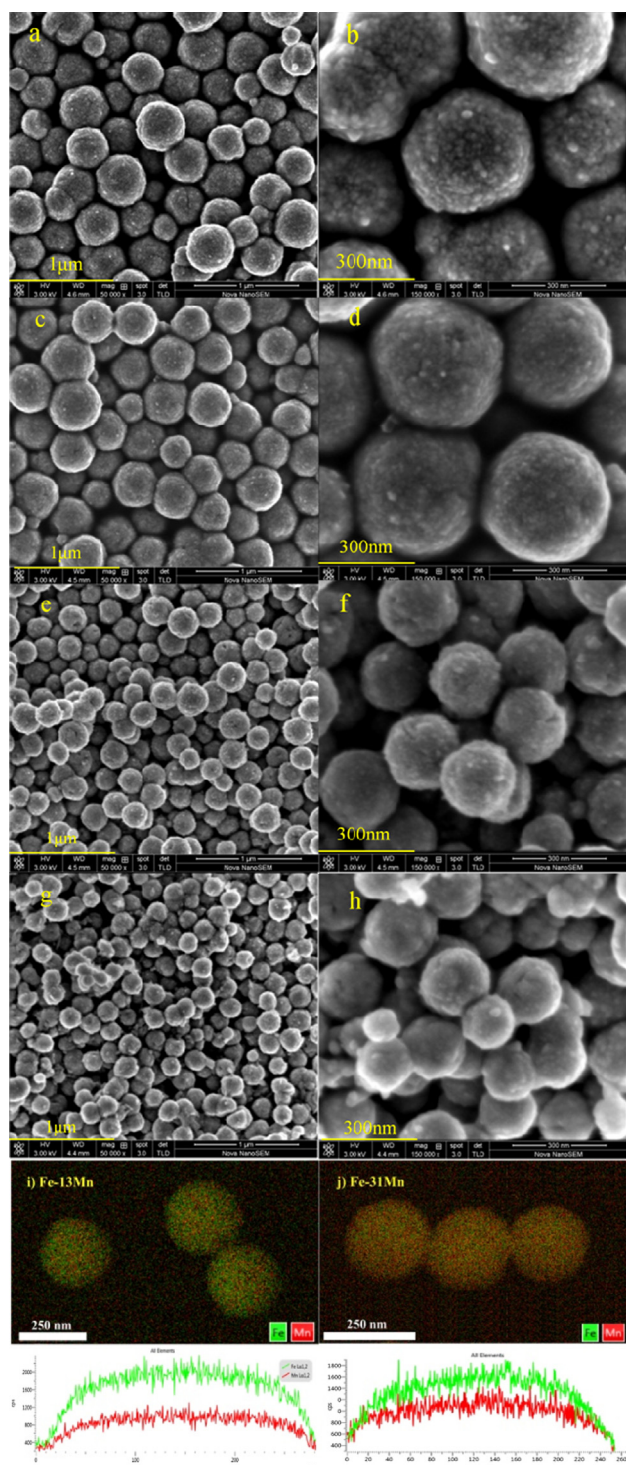


Fig. 1. SEM images of Fe-Mn microspheres. (a, b) Fe-0Mn; (c, d) Fe-2Mn; (e, f) Fe-13Mn; (g, h) Fe-31Mn. SEM-EDS mapping of the Fe-13Mn (i) and Fe-31Mn (j) microspheres. The line profiles across the sample are shown below the EDS images. Iron is presented in green, manganese in red.

become smoother, and the mean grain sizes of the microspheres decrease from 390 nm to 180 nm with the specific surface area increasing accordingly from 43 to 213 m²/g (Table 1). The FeMn microsphere with higher Mn content possesses relatively low surface energy, which results in the less extent of self-aggregation under identical solvothermal conditions [30,39]. ICP-MS and EDS analyses (Table 1) indicated that not all the Mn in precursors

was doped in the microspheres. As for Fe-2Mn, only ~2% Mn was doped into the microsphere of spinel ferrite. Further increasing the nominal Mn loading, only 13.3% and 31.4% Mn penetrated the nanocrystals of Fe-13Mn and Fe-31Mn, respectively. EDS mapping and line profiles across the sample (Fig. 1i-j) demonstrated that the as-prepared samples are solid microspheres with uniform distribution of Fe and Mn elements. Interestingly, the real Mn molar content in total metal ions of $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ is less than 33.3% in spinel MnFe_2O_4 , suggesting that the x value in the as-prepared $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ microsphere is in the range $0 < x < 1$ under such solvothermal conditions.

From the Raman spectra (Fig. 2), Fe-0Mn and Fe-2Mn both exhibit the main band at around 660 cm⁻¹ corresponding to the A_{1g} vibration mode of symmetric stretching of O atoms along Fe–O bond in Fe₃O₄ [19]. For Fe-13Mn and Fe-31Mn, the peaks at 339, 625 cm⁻¹ correspond to the E_g and A_{1g} modes of spinel MnFe_2O_4 [40], and the one at 598 cm⁻¹ could be attributed to the A_{1g} stretching mode of MnO [41], suggesting the formation of the spinel structure and the enrichment of MnO on the catalyst surface.

The diffraction peaks of Fe-0Mn, Fe-2Mn and Fe-13Mn microspheres (Fig. 3) at 18.3°, 30.1°, 35.5°, 43.1°, 53.5°, 57.0°, 62.6° could be assigned to cubic inverse spinel Fe₃O₄ (ICDD PDF#04-009-4225) and those of Fe-31Mn at 18.1°, 29.9°, 35.2°, 42.8°, 53.1°, 56.7°, 62.1° ascribed to spinel MnFe_2O_4 (ICDD PDF#01-071-4919) structure. When the Mn content increased from 2% to 13%, Fe-Mn microspheres correspond to the cubic spinel phase, and no phase related to Mn was detected. Further increasing the Mn content, the diffraction peaks become broader and less intensive, it indicates a decrease in the crystallinity of Fe-Mn microspheres. By using the Topas software from Bruker Company, the crystal size and lattice parameters of cubic ferrite were calculated and listed in Table 1. With the gradual rising of Mn, the overall diffraction peaks of Fe-Mn microspheres shift slightly toward the lower angles while the estimated lattice parameter increases from 8.39 Å of Fe-0Mn to 8.44 Å of Fe-31Mn, resulting from the larger ionic radius of Mn as compared to Fe [42–44]. Theoretically, the lattice parameters of Fe₃O₄ and MnFe_2O_4 are 8.39 and 8.51 Å, respectively [44–46]. Thus, the lattice expansion induced by Mn evidenced the formation of solid solution of spinel Fe-Mn ferrite ($\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$), corresponding well with the similar variation of lattice parameter in previous studies [44,46].

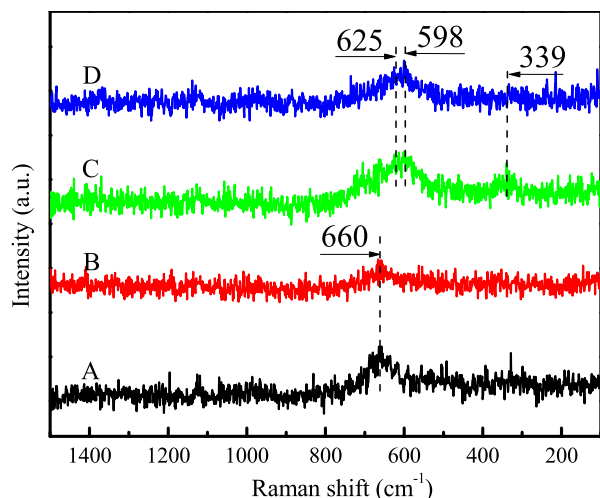
3.2. Reducibility of Fe-Mn microspheres

H₂-TPR (Fig. 4) was performed to determine the reducibility of Fe-Mn microspheres. During solvothermal synthesis of Fe-Mn microspheres, the divalent Mn salt was used as the Mn precursor, and ethylene glycol was served as solvent and reducing agent. Probably, MnO was the primary structure of Mn after the solvothermal process. As a typical manganese oxide, the reduction behavior of Mn₂O₃ were studied in a previous study [47], and the reductive evolution of Mn₂O₃ follows an order of $\text{Mn}_2\text{O}_3 \rightarrow \text{Mn}_3\text{O}_4 \rightarrow \text{MnO}$. It is noted that MnO could no longer be reduced to metallic Mn below 1000 K, consistent with the thermodynamic predictions from the Mn–O–H phase stability diagrams [48]. In this work, the reduction behavior of as-prepared manganese oxide (MnO) was also measured under H₂, and no reduction peak related to Mn was detected (Fig. 4E). The reduction of Fe-Mn samples can be consisted of into two steps: spinel $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ was firstly reduced to $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ phase below 887 K, followed by further reduction of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ to Fe⁰ and MnO at ca. 950 K. When the Mn content in the microsphere becomes low, the two reduction peaks will overlap. With an increase in the Mn content from 2% to 31%, the overlapped peaks are entirely separated as the first peak appeared below 900 K moves to lower temperatures. The reduction peak of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ broadens with the unobvious peak

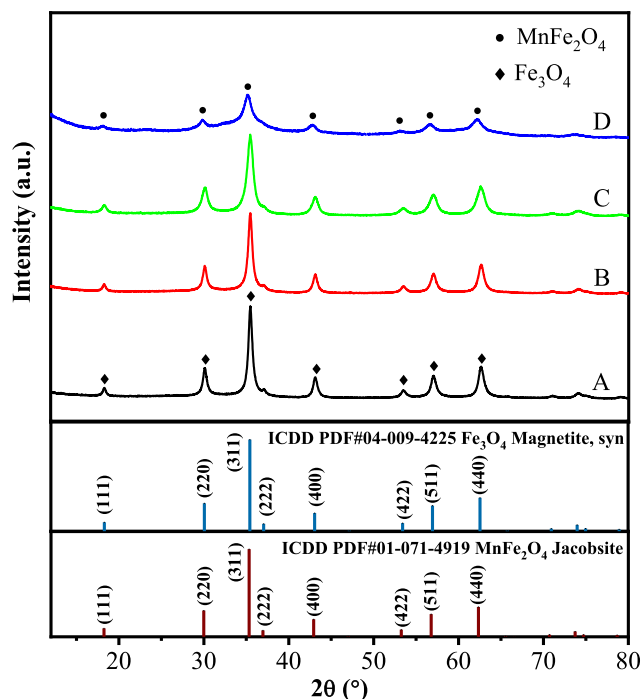
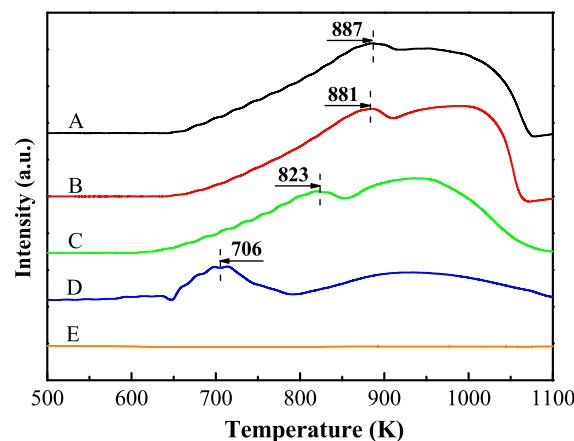
Table 1

Specific surface area, size, lattice parameters, and composition of the Fe-Mn microspheres.

Sample	S_{BET} , $\text{m}^2/\text{g}^{\text{a}}$	V_{pore} , $\text{cm}^3/\text{g}^{\text{a}}$	d_{pore} , nm^{a}	Mean size of microspheres, nm^{b}	Crystal size, nm^{c}	Lattice parameter, $\text{\AA}^{\text{d}}$	Mn content, % ^e	Mn content, % ^b
Fe-0Mn	43	0.066	6.8	390	15.8	8.39	0	0
Fe-2Mn	57	0.063	5.4	370	14.8	8.39	2.4	2.2
Fe-13Mn	120	0.074	3.9	250	10.5	8.40	13.3	10.5
Fe-31Mn	213	0.186	3.5	180	8.3	8.44	31.4	28.1

^a Determined by N_2 physisorption.^b Mean size of microspheres, calculated from SEM images; Mn content in total metal ions, determined by SEM-EDS.^c Estimated based on Scherrer equation.^d Calculated based on the highest diffraction peak with intensity >10,000 CPS in XRD patterns.^e Mn content in total metal ions, determined by ICP-MS.**Fig. 2.** Raman spectra of fresh Fe-Mn microspheres. (A) Fe-0Mn; (B) Fe-2Mn; (C) Fe-13Mn; (D) Fe-31Mn.

shift at above 900 K, suggesting that Mn promotes the reduction of $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ and stabilizes $\text{Mn}_x\text{Fe}_{1-x}\text{O}$. From quantitative analysis

**Fig. 3.** XRD patterns of Fe-Mn microspheres. (A) Fe-0Mn; (B) Fe-2Mn; (C) Fe-13Mn; (D) Fe-31Mn.**Fig. 4.** TPR profiles of Fe-Mn microspheres with different Mn contents in 10% H_2/Ar . (A) Fe-0Mn; (B) Fe-2Mn; (C) Fe-13Mn; (D) Fe-31Mn; (E) MnO.

(Fig. S1, Table S1), the decrease in the H_2 consumption is more pronounced as the Mn loading increases (Fig. S1). And compared with the required H_2 for complete reduction of the catalysts to metallic Fe, the reduction degree of Fe-13Mn and Fe-31Mn catalysts is only 89.2% and 46.9%, respectively (Table S1). The above results indicate that the reduction of the spinel ferrite is suppressed with increasing Mn loading, possibly due to the stabilization of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ phase.

To explore the structure evolution of Fe-Mn microspheres, *in situ* XRD patterns were collected over different Mn contents during reduction (Fig. 5). Almost all the diffraction peaks slightly shift to the lower angles and become more intensive and sharper as the temperature and the reduction time increases. For instance, the spinel ferrite (3 1 1) diffraction peaks for Fe-0Mn, Fe-2Mn, Fe-13Mn, and Fe-31Mn shift from 35.6° , 35.6° , 35.5° and 35.3° to 35.4° , 35.3° , 35.2° and 35.1° after reduction for 5 h, respectively, accompanied with a slight increase in the lattice spacing with the rise of temperature. In addition, the sharper diffraction peaks indicate the growth of nanocrystals at high-temperature reductive environment. Being consistent with *ex situ* XRD patterns of Fe-Mn microspheres, the addition of Mn expands the lattice parameter, as indicated by a downshift in the overall diffraction peaks of Fe-Mn microspheres.

More specifically, as shown in Fig. 5a, the diffraction peaks of Fe_3O_4 (ICDD PDF#04-009-4225) remained unchanged below 523 K during reduction. The intensity of (2 2 2) facet of Fe_3O_4 at ca. 37.1° became distinct at 523 K. After holding at 623 K for 1 h, two peaks at 44.5° and 64.7° associated with the Fe^0 phase (ICDD PDF#04-002-8917) appeared and became more intensive with time. Therefore, two phases, namely cubic Fe^0 and cubic Fe_3O_4 , were detected for Fe-0Mn after reduction. Table 2 summarizes the phase composition of the reduced Fe-Mn samples through

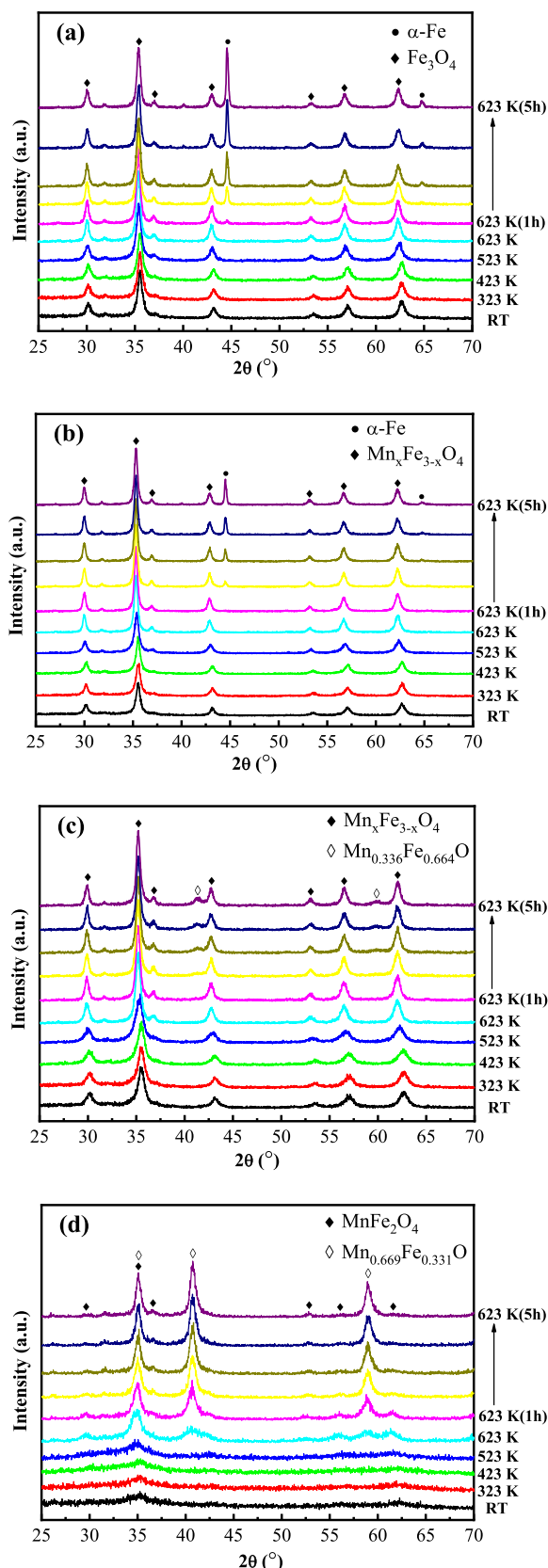


Fig. 5. In situ XRD patterns of Fe-Mn microspheres during reduction under 5% H₂/Ar. (a) Fe-0Mn; (b) Fe-2Mn; (c) Fe-13Mn; (d) Fe-31Mn.

Rietveld refinement. The content of Fe⁰ decreases for Fe-2Mn (10.7%) in comparison with Fe₃O₄ (22.8%). Apparently, the pres-

Table 2

Phase composition of reduced Fe-Mn microspheres determined from Rietveld refinement.^a

Sample	Weight fraction of crystalline phases (wt. %)		
	Mn _x Fe _{3-x} O ₄	α-Fe	Mn _x Fe _{1-x} O
Fe-0Mn	77.2 ^b	22.8	— ^c
Fe-2Mn	89.3	10.7	0.3
Fe-13Mn	73.2	—	26.8
Fe-31Mn	2.4	—	97.6

^a Reduction conditions: 623 K, 1 atm for 5 h under H₂ with a ramping rate of 5 K/min.

^b Fe₃O₄.

^c “—” not detected.

ence of Mn could suppress the reduction of Fe₃O₄ to Fe⁰. For Fe-13Mn (Fig. 4c), no peak associated with the cubic Fe⁰ was detected during reduction in H₂. However, two new diffraction peaks at 41.4° and 59.9° appeared, being attributed to the cubic Mn_{0.336}Fe_{0.664}O phase (ICDD PDF#01-077-2358) [33]. The presence of Mn_xFe_{1-x}O during reduction, generally as the intermediate phase, may arise from the stabilization effect of manganese. After H₂ treatment for 5 h, main crystal phase Mn_xFe_{3-x}O₄ (73.2%) and Mn_xFe_{1-x}O (26.8%) co-exist in the reduced Fe-13Mn. For Fe-31Mn, 97.6% Mn_{0.669}Fe_{0.331}O (ICDD PDF#01-077-2360) and a small part of MnFe₂O₄ (2.4%) was measured after reduction. No metallic Fe⁰ was detected by XRD for both samples after reduction.

3.3. Structure of the spent Fe-Mn microspheres

The Fe2p XPS spectra of spent Fe-0Mn and Fe-13Mn catalysts are shown in Fig. 6. The peaks at 710.9 eV and 724.5 eV can be assigned to the spin-coupled doublet for Fe 2p_{3/2} and Fe 2p_{1/2} of α-Fe₂O₃ [49], originating from the passivation of spent catalysts by 1% O₂/N₂. The inconspicuous shoulders at 707.5 eV and 720.6 eV can be attributed to the spin-coupled doublet for Fe 2p_{3/2} and Fe 2p_{1/2} of χ-Fe₅C₂ [50]. Before sputtering, the surface of spent catalysts is composed of α-Fe₂O₃ with a small amount of χ-Fe₅C₂. After sputtering for 1 or 2 min, the BE of Fe 2p_{3/2} and Fe 2p_{1/2} at 707.5 eV and 720.6 eV, corresponding to χ-Fe₅C₂ phase, were observed. However, concerning the spent Fe-13Mn, carbon occupied the whole surface and almost no signal in Fe spectrum was detected before sputtering. After sputtering for 2 mins, only a weak peak at 707.5 eV assigned to χ-Fe₅C₂ can be observed. Notable difference of the surface structure was observed between spent Mn-doped and pure Fe₃O₄. The deposited carbon of spent Fe-13Mn was much thicker than that of unpromoted catalyst, indicating the improvement of CO dissociation by doping MnO_x.

Raman spectra of spent Fe-Mn catalysts are shown in Fig. 7. After the FTO reaction, spent Fe-0Mn possessed specific bands of α-Fe₂O₃ (A_{1g} mode at 215 cm⁻¹, three E_g modes at 277 cm⁻¹, 398 cm⁻¹, 612 cm⁻¹) [51], carbonaceous species at ca. 1350 cm⁻¹ (D band) and 1590 cm⁻¹ (G band) [52]. Like XPS results, the bands of α-Fe₂O₃ came from the passivation of spent catalysts. D band is usually assigned to the C=C stretching in disordered amorphous carbonaceous species, while the G band, attributed to the tangential E_{2g} phonon of in-plane stretching vibration of the C=C bond, is related to the graphitic carbon layers [52]. With the addition of Mn, the intensity of the vibration mode of Fe₂O₃ decreases dramatically, probably due to the increase in the thickness of deposited carbon layers. As for Fe-13Mn, almost no vibrational frequency of iron oxide was detected (C in Fig. 7). In contrast, the broad band in the range 595–625 cm⁻¹ could be attributed to the symmetric Mn-O stretching vibrational A_{1g} mode of MnO [41] and manganese ferrite (MnFe₂O₄) [40] in the spent Fe-31Mn. Further addition of

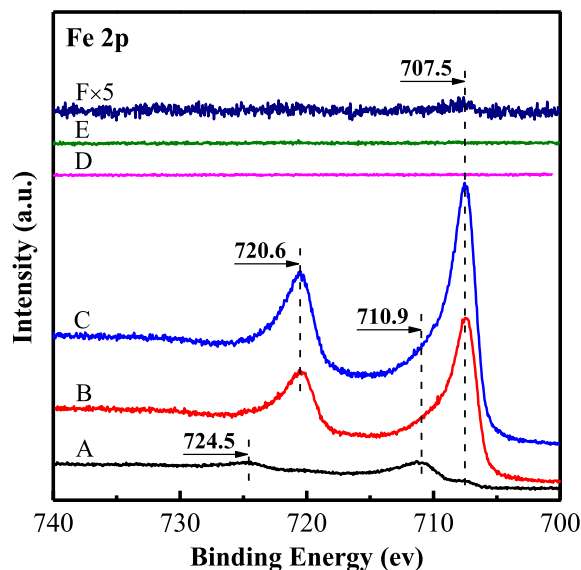


Fig. 6. Fe 2p XPS spectra of the spent Fe-Mn microspheres. An Ar⁺ ion gun operated with a beam voltage of 3 kV in a 2 × 2 mm² was used to acquire XPS depth profiles. (A) Fe-0Mn; (B) Fe-0Mn sputtered for 1 min; (C) Fe-0Mn sputtered for 2 min; (D) Fe-13Mn; (E) Fe-13Mn sputtered for 1 min; (F) Fe-13Mn sputtered for 2 min.

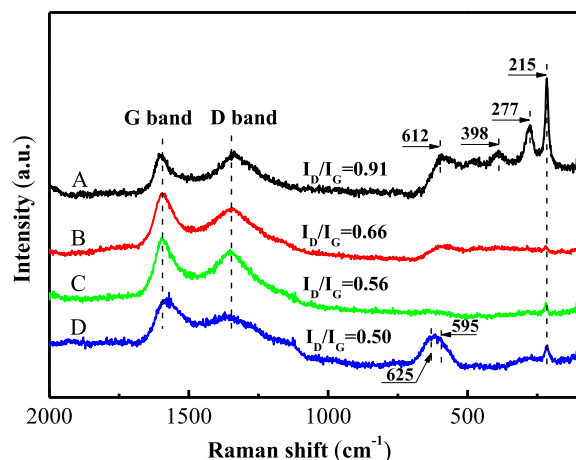


Fig. 7. Raman spectra of spent Fe-Mn microspheres. (A) Fe-0Mn; (B) Fe-2Mn; (C) Fe-13Mn; (D) Fe-31Mn.

Mn may lead to the formation of Mn-doped spinel ferrite and prevent the iron oxide from carburization.

Generally, the integrated intensity ratio of D to G band (I_D/I_G) indicates the defect concentration and the degree of graphitization [53]. Based on our calculations, the I_D/I_G ratios are 0.91, 0.66, 0.56 and 0.50 for the spent Fe-0Mn, Fe-2Mn, Fe-13Mn and Fe-31Mn catalysts, indicating that the graphitization degree of carbonaceous species increases with the increasing amount of Mn. We assume that the doping of Mn may enhance the adsorption and dissociation of CO on the catalyst surface, resulting in the higher degree of graphitization. Also, it should be noted that the full widths at half maximum (FWHM) of D and G band on Fe-31Mn are the largest among all the spent samples, manifesting the poorest crystallinity of deposited carbon on the surface of spent Fe-31Mn. Regarding the strong Mn-O vibrational mode, MnO could be enriched on the spent Fe-31Mn catalyst.

XRD was further employed to investigate the crystal structure and composition of the spent catalysts (Fig. 8). Evidently, the crystal structures of spent Fe-0Mn and Fe-2Mn catalysts are rather

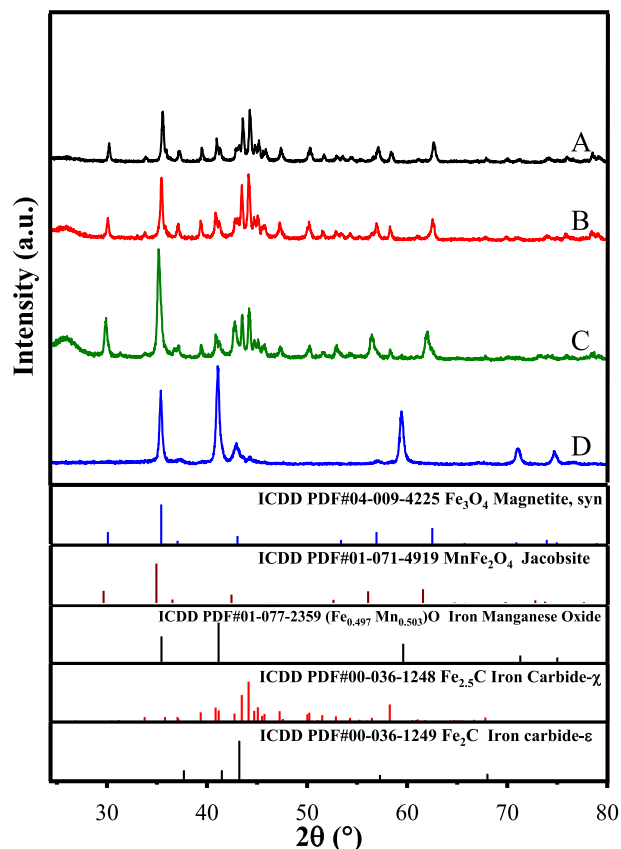


Fig. 8. XRD patterns of spent Fe-Mn microspheres. (A) Fe-0Mn; (B) Fe-2Mn; (C) Fe-13Mn; (D) Fe-31Mn.

similar. The diffraction peaks at 30.1°, 35.4°, 37.1°, 43.1°, 53.4°, 56.9°, 62.5° and 73.9° represent the (2 2 0), (3 1 1), (2 2 2), (4 0 0), (4 2 2), (5 1 1), (4 4 0), (5 3 3) planes of cubic Fe₃O₄ (ICDD PDF#04-009-4225), respectively. The other characteristic peaks at 39.4°, 40.9°, 41.2°, 43.5°, 44.2°, 44.7°, 45.1°, 47.3°, 50.2° and 58.3° correspond well with the (0 2 0), (1 1 -2), (2 0 2), (0 2 1), (5 1 0), (4 0 -2), (3 1 -2), (2 2 1), (3 1 2), (2 2 2) planes of χ -Fe₅C₂ (ICDD PDF#00-036-1248). With an increase in the Mn content, the characteristic peaks of Fe₃O₄ shifted toward the lower diffraction angles, indicating the formation of spinel-like Mn_xFe_{3-x}O₄ phase. Spinel ferrites (Fe₃O₄/Mn_xFe_{3-x}O₄) and χ -Fe₅C₂ are the main phases of spent Fe-Mn samples with the Mn content below 13%. Moreover, it is noteworthy that a new phase ϵ -Fe₂C (ICDD PDF#00-036-1249), at 37.7°, 41.5°, 43.2°, 57.3°, 68.0° corresponding to the (1 0 0), (0 0 2), (1 0 1), (1 0 -2), (1 1 0) planes, appeared in the spent Fe-13Mn and became the dominant carbide phase in spent Fe-31Mn. Another Mn_{0.503}Fe_{0.497}O phase at 35.4°, 41.2°, 59.6°, 71.3°, 75.0° indexable to (1 1 1), (2 0 0), (2 2 0), (3 1 1), (2 2 2) planes (ICDD PDF#01-077-2359) was detected as the main phase in the spent Fe-31Mn. In addition, the weak broad diffraction at $2\theta = 26^\circ$ can be attributed to the amorphous deposited carbon produced during the FTO reaction, and the carbon intensity increases with the Mn content below 13%. However, subtle deposited carbon was found in the spent Fe-31Mn.

Table 3 displays the quantitative phase composition in spent Fe-Mn microspheres determined by Rietveld refinement of XRD patterns. The spent Fe-2Mn shows similar phase composition with the spent Fe-0Mn catalyst, it means that the addition of a small amount of Mn (below 2%) has little impact on the long-range structure of Fe microsphere. For spent Fe-0Mn, more than 70% of oxides were transformed into χ -Fe₅C₂. Ordered crystal of ϵ -Fe₂C began to

Table 3

Phase composition of the spent samples determined from Rietveld refinement.

Spent catalysts	Weight fraction of crystalline phases (wt. %)					
	Mn _x Fe _{3-x} O ₄	Mn _x Fe _{1-x} O	Sum of oxides	χ-Fe ₅ C ₂	ε-Fe ₂ C	Sum of carbides
Fe-0Mn	29.2 ^a	– ^b	29.2	70.8	–	70.8
Fe-2Mn	28.5	–	28.5	68.9	2.6	71.5
Fe-13Mn	45.9	3.6	49.5	42.3	8.2	50.5
Fe-31Mn	0.5	76.1	76.6	7.7	15.7	23.4

^a Fe₃O₄.^b “–” not detected.

appear in the spent Fe-2Mn. Moreover, with increasing the Mn content from 2% to 31%, the content of ε-Fe₂C increased from 2.6% to 15.7%, whereas the content of χ-Fe₅C₂ decreases from 68.9% to 7.7%.

Notably, Mössbauer spectra also indicates the ε-Fe₂C/χ-Fe₅C₂ ratio rises significantly from 0.56 to 2.58 with the increase of Mn (Fig. 9, Table S2–S3), highly consistent with the previous studies [55,59]. Moreover, along with the total oxide content increasing from ca. 29.2% to 76.6%, the main oxidic phase transforms from spinel ferrites (Fe₃O₄/Mn_xFe_{3-x}O₄) to Mn_xFe_{1-x}O. As the Mn content reaches 31%, the Fe²⁺ seems to be stabilized in Mn_xFe_{1-x}O during reaction and the carburization of magnetite to iron carbides is remarkably suppressed, in accordance with the previous work in which pure Fe₃O₄ was mainly carburized to χ-Fe₅C₂, and Mn-promoted samples exhibited less carbides contents after exposure to syngas [32].

3.4. Catalyst performance of Fe-Mn microspheres

The FTO performance over Fe-Mn microspheres (533 K, 2.0 MPa, and H₂/CO = 1) was evaluated, as shown in Fig. 10 and Table S4. Although the percentage of iron decreases, the incremental surface area with reduced particle size probably compensates the reduction in molar content of iron due to the increasing loading of Mn, leading to similar or even slightly higher CO conversion. However, the conversion of CO dropped to 3.3% with further addition of Mn (31%), likely due to the combined effect of reduced Fe loading and content of active carbides. Analogously, the iron time yield (FTY) rose first to a summit of 16.9 μmol_{CO}·g_{Fe}⁻¹·s⁻¹ over Fe-13Mn and then dropped to 10.9 μmol_{CO}·g_{Fe}⁻¹·s⁻¹ over Fe-31Mn.

Regarding the hydrocarbon distribution over Fe-Mn microspheres, a volcanic evolution of the selectivity to lower olefins and O/P ratio vs the Mn content was observed. With increasing the Mn content from 0 to 13%, the selectivity to C₂₋₄ olefins and CO₂ increased from 33.0% to 54.4% and from 16.7% to 25.2%, respectively. By taking a closer look at the olefinic composition in the C₂₋₄ range (Table S4), we can find that the most abundant olefin is propylene accounting for around 50% of C₂₋₄ olefins in all the samples. As the Mn loading increases from 0 to 13%, the ethylene selectivity increases from 6.5% to 13.6%, and the proportion of ethylene in the lower olefins rises from ~20% to 25%. Further increasing the Mn loading to 31% resulted in a slight drop in selectivities to lower olefins and CO₂, but a little higher CH₄ selectivity. Additionally, the O/P ratio of 3.6 over Fe-13Mn is about twice higher than that of Fe-0Mn and slightly decreases to 3.0 with the 31% Mn addition. Among these catalysts, Fe-13Mn showed the highest STY of C₂₋₄ olefins (73.0 g·kg_{cat}⁻¹·h⁻¹), the highest FTY of 16.9 μmol_{CO}·g_{Fe}⁻¹·s⁻¹ as well as the highest selectivity toward lower olefins (54.4%) along with relatively low selectivity to CH₄ (17.0%), indicating that the addition of appropriate amount of Mn is beneficial to promote the formation of lower olefins.

To further explore the influence of H₂/CO ratios and the reaction temperatures on the FTO performance, the performance tests at different H₂/CO ratios of 0.5 to 3.0 (Fig. S2, Table S5) and

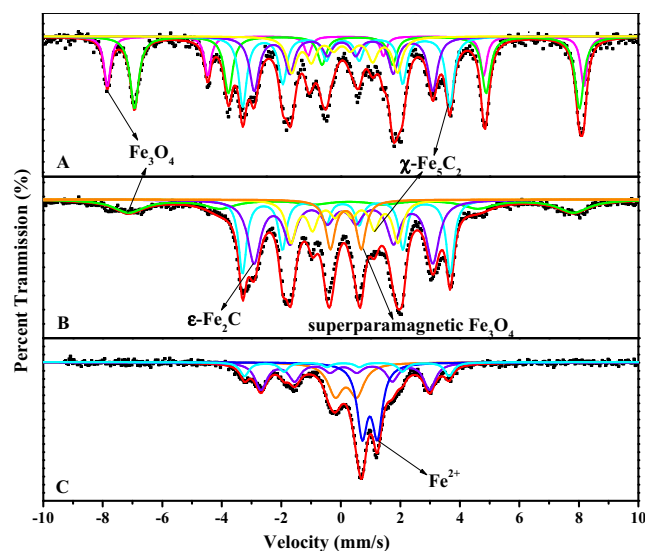


Fig. 9. ⁵⁷Fe Mössbauer spectra of the spent Fe-Mn microspheres. (A) Fe-0Mn; (B) Fe-13Mn; (C) Fe-31Mn.

temperature-scanning experiments in the range 493–613 K (Fig. S3, Table S6) were carried out over Fe-13Mn catalyst. From Table S5, as the H₂/CO ratios increasing from 0.5 to 3.0, CO conversion rises from 4.2% to 11.8% and CH₄ selectivity increases from 13.2 to 29.6%. In contrast, the selectivities to lower-olefin, C₅₊ and CO₂ drop from 60.2% to 40.3%, from 32.7 to 17.9%, from 15.1 to 8.5%, respectively. It can be seen that lowering the H₂/CO ratio benefits the C₂₋₄ olefin selectivity. In addition, both the FTY and STY show a volcanic trend and reach a maximum at H₂/CO = 1 (Fig. S2), manifesting that H₂/CO ratio of 1 is the most favorable for the production of lower olefins. From the temperature-scanning experiments, CO conversion and FTY of Fe-13Mn shows a rapid growth with increasing temperature ranging from 493 to 613 K (Fig. S3, Table S6). Apparently, the selectivities to CH₄ and CO₂ are remarkably improved with increasing temperature, whereas high temperature is strongly detrimental to the C₅₊ selectivity and olefin/paraffin ratio. As reaction temperature increases, the rapidly growing activity, along with reduced O/P ratio, renders the C₂₋₄ olefin selectivity a volcano-like shape, ultimately leading to an approximately linear STY of C₂₋₄ olefins. The highest selectivity to C₂₋₄ olefins is obtained at 533 K, however the highest lower-olefin STY of 207.0 g·kg_{cat}⁻¹·h⁻¹ is achieved at 613 K accompanied with a large amount of undesired CH₄ and CO₂.

4. Discussion

4.1. The Mn doping effect on the initial structure of spinel ferrite

In this solvothermal system, the sodium acetate acts as a coupling agent providing a slightly alkaline environment. During

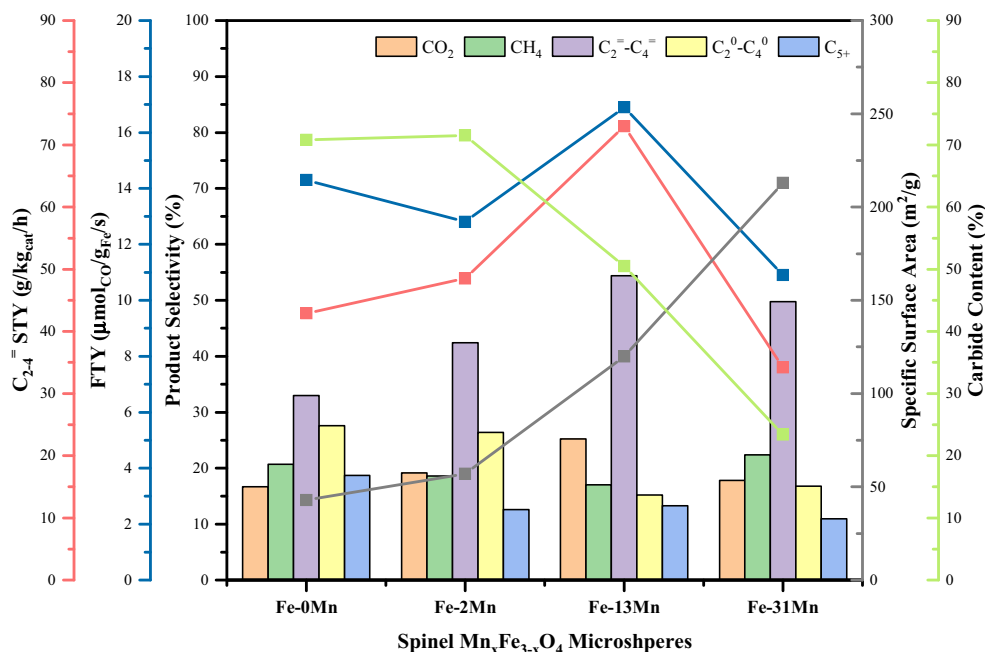


Fig. 10. Relationship of FTO performance, specific surface area, and carbide content of 20-h spinel $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ catalysts with different Mn contents^{a,b,c,d}. ^aReaction conditions: 25 mg catalyst diluted with SiC (1:3), 533 K, 2.0 MPa, $\text{H}_2/\text{CO} = 1$, GHSV = 4000 h^{-1} , TOS = 20 h. ^bThe hydrocarbon distribution is normalized on a carbon basis with the exclusion of CO_2 . ^cIron time yield (FTY) = mol of CO converted to hydrocarbons (excluding CO_2) per time (s) per weight of iron (g). ^dSTY = space time yield.

preparation, the acetate groups adsorbed on the surface of the magnetic nanocrystals could result in the formation of electrostatic repelling forces between the neighboring nanocrystals, leading to the uniform and monodisperse nanocrystals produced in the solution. On the contrary, due to the high surface energy of the magnetic nanocrystals, the nanocrystals are prone to aggregate together to reduce the surface energy [54]. The degree of aggregation of microspheres depends on the balance of these two opposing forces. With an increase in the Mn/Fe ratio during crystal growth, the doping of Mn may reduce the surface energy of nanocrystals [30] and weaken the tendency of aggregation of magnetic nanocrystals, leading to the decrease in the crystallinity and the grain size of microspheres, as evidenced by SEM images and XRD patterns (Fig. 1, Fig. 3, Table 1). In addition, the lattice distortion is induced mainly by the mismatch of Mn since the ionic radius of Mn^{2+} (0.80 Å) is larger than Fe^{2+} (0.74 Å) and Fe^{3+} (0.64 Å) [42,43]. On the other hand, the reduction of particle size is also responsible for the lattice distortion in the Mn-doped ferrite microspheres [55]. XRD patterns reveal that the crystal lattice spacing of cubic Fe_3O_4 expanded with the increasing of Mn content, manifesting that the nonstoichiometric $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ spinel ferrites were formed [44,46].

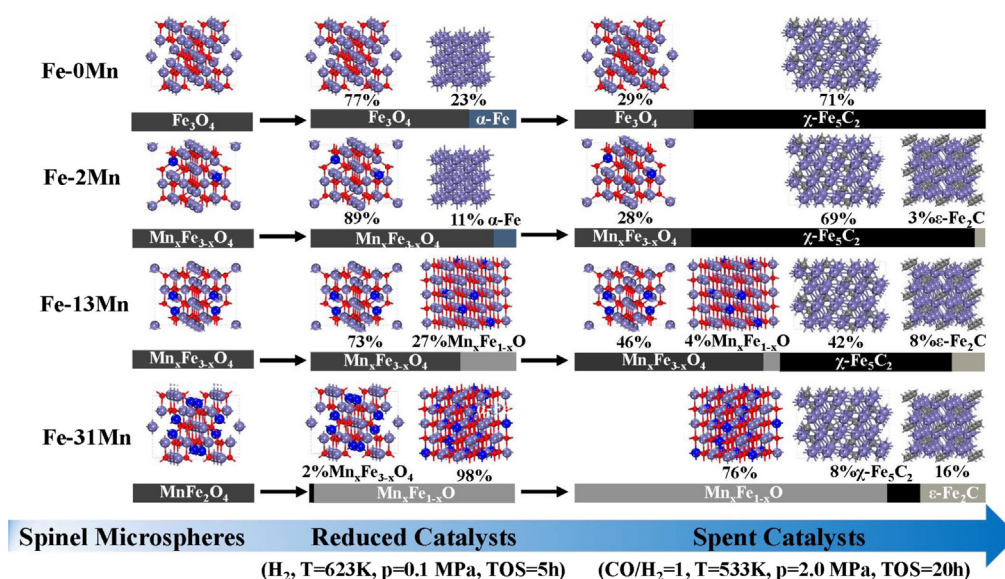
The cation distribution in spinel structures is related to the crystal field stabilization energies (CFSE) of the constituent transition metals. It has already been established that the spinel-like Fe_3O_4 has two ions (Fe) positions at the tetrahedral A site and the octahedral B site surrounded by oxygen. Fe ions located at the A site are trivalent cations, and those at the B site are a mixed-valence state between di- and trivalent cations [56]. Mn ions in $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ ($0 < x \leq 1$) are mainly divalent, and the majority of Mn^{2+} ions occupied the tetrahedral A sites, while Fe ions are mainly trivalent, and the majority of Fe^{3+} occupied the octahedral B sites [57]. However, Fe ions in $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ ($1 < x < 3$) are mostly trivalent and the majority of Fe^{3+} ions located on the octahedral B sites, while Mn ions are mixed-valent with Mn^{2+} at the tetrahedral A sites and Mn^{3+} ions at the octahedral B sites [58]. Considering the ion radius of Mn^{2+} (0.80 Å) and substituted Fe^{3+} (0.64 Å),

Yamanaka found that the lattice constant sharply rose from Fe_3O_4 to stoichiometric MnFe_2O_4 [44]. With further increasing Mn content to 66%, i.e., Mn_2FeO_4 , only a subtle increase in the lattice constant was measured because there is little difference for the ion radius of Mn^{3+} (0.66 Å) and Fe^{3+} (0.64 Å) at the octahedral position. ICP-MS, XRD and EDS analyses demonstrated the as-prepared Fe-Mn microspheres are $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ with x value in the range of $0 < x < 1$ during the solvothermal process (Table 1). Takaobushi et al. [57] validated that Mn^{2+} substitutes the A sites in $\text{Fe}_{3-x}\text{Mn}_x\text{O}_4$ along with the reduction in Fe^{2+} and has little impact on the electronic state of Fe 2p core structure of $\text{Fe}_{3-x}\text{Mn}_x\text{O}_4$ spinel oxide, thus suggesting the negligible electronic effect of Mn in the spinel structure $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ ($0 < x \leq 1$).

4.2. Structure evolution during activation and FTO reaction

According to H_2 -TPR profiles, a small amount of Mn can lower the reduction temperature of spinel ferrites ($\text{Fe}_3\text{O}_4/\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$) to wüstite ($\text{FeO}/\text{Mn}_x\text{Fe}_{1-x}\text{O}$). Furthermore, the reduction peak of $\text{Fe}_3\text{O}_4 \rightarrow \text{FeO}$ and $\text{FeO} \rightarrow \text{Fe}$ was gradually separated with the addition of Mn, and divided entirely into two clear-cut peaks for Fe-31Mn; it indicates the stabilization of wüstite by Mn(II). Sano et al. proposed that anion repulsion would become weaker with increasing the Mn(II) content, thus leads to the decrease in the Gibbs energy of Mn(II)-wüstite ($\text{Mn}_x\text{Fe}_{1-x}\text{O}$) formation [59], due to the expansion of the lattice spacing between oxide ions. In addition, it was also reported that the excess of Mn may facilitate the formation of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ solid solution, and meanwhile suppressed the phase transformation of FeO to metallic Fe [60], which is in good agreement with *in situ* XRD patterns and the conspicuously broadened reduction of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ in H_2 -TPR profiles. Probably, the difference between the Gibbs free energy of formation of Mn(II)-wüstite and Mn(II)-ferrite is too small to compensate the Gibbs energy of α -Fe formation from Mn(II)-wüstite [59].

Scheme 1 illustrates the phase evolution during FTS over Fe-Mn microspheres. For as-prepared Fe-0Mn and Fe-2Mn, spinel ferrites and α -Fe were the main phases after reduction in H_2 , and the



Scheme 1. The phase evolution during activation and FTO reaction over spinel Fe-Mn microspheres based on the Rietveld refinement of XRD patterns (Fe, light purple; Mn, bright blue; O, red; C, grey).

ferritic phase fraction increased from 77% (Fe_3O_4) to 89% ($\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$) along with a decrease in $\alpha\text{-Fe}$ content from 23% to 11% over Fe-2Mn compared with Fe-0Mn (Table 3). The corresponding spent samples were composed of $\sim 70\%$ $\chi\text{-Fe}_5\text{C}_2$ and $\sim 30\%$ spinel ferrites ($\text{Fe}_3\text{O}_4/\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$) based on Rietveld refinement of XRD patterns. However, the quantitative analyses from Mössbauer spectra showed that spent Fe-0Mn consisted of 32.3% Fe_3O_4 , 32.2% $\chi\text{-Fe}_5\text{C}_2$, and 35.5% $\varepsilon\text{-Fe}_2\text{C}$ that was not discriminated by XRD (Table S2). Wei et al. also observed the invisible $\varepsilon\text{-Fe}_2\text{C}$ in the XRD patterns and attributed this to the fact that the small crystallite size of $\varepsilon\text{-Fe}_2\text{C}$ was below the detection limit of XRD [61]. With the addition of Mn, new $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ phases were observed while $\alpha\text{-Fe}$ disappeared in the reduced Fe-13Mn and Fe-31Mn. The content of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ phase increased from 27% to 98% with the actual Mn content from 13% to 31%. After the FTO reaction, the Mn-doped spinel ferrite and Mn(II)-wüstite partly transformed into $\chi\text{-Fe}_5\text{C}_2$ (42%) and $\varepsilon\text{-Fe}_2\text{C}$ (8%) with 46% $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ and 4% $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ left in the bulk phase of Fe-13Mn. Curve fitting of Mössbauer spectra displayed more carbide content (74.2%) in the spent Fe-13Mn, especially $\varepsilon\text{-Fe}_2\text{C}$ (49.9%). For the spent Fe-31Mn, $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ (76%) prevailed over iron carbides which were composed of 8% $\chi\text{-Fe}_5\text{C}_2$ and 16% $\varepsilon\text{-Fe}_2\text{C}$. Similarly, the spent Fe-31Mn comprises 12.1% $\chi\text{-Fe}_5\text{C}_2$, 31.2% $\varepsilon\text{-Fe}_2\text{C}$ and 56.7% superparamagnetic oxides with small sizes according to Mössbauer results. In line with our iron phase transition in syngas, Mogorosi et al. deduced that $\alpha\text{-Fe}$ was mostly transformed into $\chi\text{-Fe}_5\text{C}_2$, whereas FeO was the main source for $\varepsilon\text{-Fe}_2\text{C}$ during FTS [62].

Clearly, with the gradual addition of Mn, the reduction of Fe_3O_4 to $\alpha\text{-Fe}$ could be restrained, and the formation of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ was facilitated. In the reduced Fe-13Mn, $\alpha\text{-Fe}$ was not detected; Fe atoms in Fe-31Mn were stabilized in the $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ solid solution after reduction. When only a small amount of Mn was doped, the main phases are spinel ferrites ($\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$) and $\alpha\text{-Fe}$ in the reduced catalysts and a high proportion of them could be converted to $\chi\text{-Fe}_5\text{C}_2$ easily during the reaction. After the FTO reaction, it should be noted that the carbide content declined markedly accompanied by a rise of oxide content (especially for $\text{Mn}_x\text{Fe}_{1-x}\text{O}$) in the spent samples with further increase in the Mn content, probably due to the decreased Gibbs energy of Mn(II)-wüstite formation induced by the expansion of the lattice spacing [59].

The relative content of $\varepsilon\text{-Fe}_2\text{C}$ in iron carbides increased with the rising of the Mn content based on the XRD and Mössbauer fitting (Tables 3, S2, S3). Generally, a theory of a sequence of carbon depletion during FTS has been proposed [2,63], in which the pseudo-hexagonal iron carbide ($\varepsilon\text{-Fe}_{2.2}\text{C}$, $\text{P6}_3/\text{mmc}$) firstly transforms to Eckström-Adcock carbide (Fe_7C_3 , $\text{P6}_3/\text{mc}$), then to Hägg carbide ($\chi\text{-Fe}_5\text{C}_2$, $\text{C12}/\text{c1}$), hexagonal ε -cementite ($\varepsilon\text{-Fe}_3\text{C}$, $\text{P6}_3/\text{mc}$), and finally to cementite ($\theta\text{-Fe}_3\text{C}$, Pnma). It agrees with the stability order of iron carbides as: $\text{Fe}_2\text{C} < \text{Fe}_{2.2-2.4}\text{C} < \text{Fe}_5\text{C}_2 < \text{Fe}_3\text{C}$ under typical FTS conditions (low carbon chemical potential μ_{C}) [64,65]. Herein, the rise in the relative content of $\varepsilon\text{-Fe}_2\text{C}$ in the spent Fe-Mn microspheres demonstrates the carbide phase transformation is impeded with an increase in the Mn content.

It has been verified that the carburization ability depends mainly on the carbon content of the gas environments (partial pressure, H_2/CO ratio) and the temperature [20,66]. Specifically, the higher carbon content of carbon-containing gases, the higher the carburization ability. Also, higher temperature, lower pressure and higher H_2/CO ratio can suppress carburization ability and retard carbon deposition. It should be mentioned that Mn is widely used as an austenite stabilizer in the production of austenite steel [67]. In good coincidence with the previous work [44,68], the substitutional solution between Fe and Mn expands the lattice constant as evidenced by XRD (Table 1), probably giving rise to the stabilization of the interstitial dissolved carbon atoms by reducing the activation energy of carbon diffusion [69,70]. Besides, Oda et al. have found that the Mn-C interaction in the face-centered cubic $\gamma\text{-Fe-Mn-C}$ is strong, possibly accounting for increased C solubility and the formation of Mn-doped iron carbides [71]. Therefore, we inferred that the expansion of the lattice parameter could reduce the Gibbs formation energy of the Mn-doped wüstite while lowering the activation energy of carbon diffusion, ultimately leading to the stabilization of the Mn(II)-wüstite during reduction and the suppression of the carbide phase transformation during the FTO reaction.

4.3. Structure-performance relationship

Based on Scheme 1, the foregoing H_2 -TPR and XRD experiments have shown that as the Mn loading increases, the reduction of Fe is significantly suppressed, which subsequently slows the phase

transition of the reduced metallic Fe and Fe oxides to Fe carbides under a syngas atmosphere. Also, according to the above discussions, the lattice spacing expansion induced by Mn improves the stability of $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ and $\varepsilon\text{-Fe}_2\text{C}$ during reduction and reaction processes. Generally, the catalytic activities were essentially determined by the active phases (i.e., iron carbides) on the surface of the catalysts. For the bulk Fe-Mn microspheres, the activity depends on the surface area dominated by the size and pore structure of microspheres, and the proportion of iron carbides (e.g., $\chi\text{-Fe}_5\text{C}_2$, $\varepsilon\text{-Fe}_2\text{C}$). With the addition of Mn, the surface area of Mn-doped ferrite microspheres increases as the particle size gradually decreases, and meanwhile, the content of active carbides decreases because of declining reducibility and the lattice expansion of spinel ferrite. Thus, the activity and FTY exhibit volcano curves and reach a summit over Fe-13Mn with 7.1% CO conversion and FTY of $16.9 \mu\text{mol}_{\text{CO}} \cdot \text{g}_{\text{Fe}}^{-1} \cdot \text{s}^{-1}$ (Fig. 10, Table S4).

On the other hand, since Mn oxide shows more basic than Fe oxides [33], extensive studies have reported that Mn could improve the surface basicity of Fe catalysts and enhance the CO dissociation and the selectivities to light olefins and C_{5+} hydrocarbons while inhibiting CH_4 formation [33,72,73]. The enhanced CO adsorption and dissociation over Fe-13Mn is also reflected in the severe carbon deposition and more graphitized carbon evidenced by the spectra of XPS and Raman and XRD patterns of spent Fe-Mn microspheres (Figs. 6–8). Probably due to a low surface energy of MnO [30], Fe catalyst has found to be covered by MnO even at the low manganese content, and the enrichment is more severe with the excess of Mn [32,74,75], it is responsible for the sharp decline in activity. Recent work showed that the surface-accumulated MnO increased the abundance of adsorbed CO and facilitates the cleavage of C–O bond through Lewis acid-base interactions [76].

About the product distribution, the selectivity to C_{5+} hydrocarbons decreased in some extent in the presence of Mn, it is probably due to the inhibition of the carbide formation during the FTO reaction. With increasing the Mn content from 0 to 13%, the olefin selectivity, O/P ratio and CO_2 selectivity increase remarkably accompanied by a drop in CH_4 selectivity, indicating that moderate amount of Mn weakens the hydrogenation of carbonaceous intermediates while enhancing the capability for CO adsorption and subsequent dissociation. In addition, the strong surface basicity could promote the olefin desorption and hinder the readsorption and secondary reactions of olefins, thereby possibly contributing to the selectivity to lower olefins. As mentioned above, the addition of Mn facilitates the formation of $\varepsilon\text{-Fe}_2\text{C}$ during reaction by expansion of the lattice spacing of $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$. Based on the Mössbauer analysis (Tables 3, S3), it appears that with the rise of the $\varepsilon\text{-Fe}_2\text{C}/\chi\text{-Fe}_5\text{C}_2$ ratio in the Fe-13Mn phase, the olefin selectivity and O/P ratio increase compared with Fe-0Mn and remain stable with the further augment of $\varepsilon\text{-Fe}_2\text{C}/\chi\text{-Fe}_5\text{C}_2$ ratio. Here, we speculated that, apart from the enhanced surface basicity, an appropriate proportion of $\varepsilon\text{-Fe}_2\text{C}/\chi\text{-Fe}_5\text{C}_2$ and manganese ferrite may jointly boost the selective formation of lower olefins with a small amount of Mn, which needs further confirm in the future. Continuous increase the Mn loading to 31% resulted in a slight drop in selectivities to lower olefins, CO_2 and O/P ratio, but a little higher CH_4 selectivity, possibly caused by the limited transformation of iron carbides and the enrichment of MnO. We posited that the Mn-doped spinel ferrite with Mn/Fe ratio close to the stoichiometric of MnFe_2O_4 robustly stabilized the $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ phase and reduced the formation of Fe_3O_4 , which is known as the active phase of water-gas-shift (WGS) reaction.

We believe that profound insights into the role of doping metals in the preparation, reduction, and reaction of spinel ferrite microspheres could pave the way for deep understanding the structure-

performance relationship and rational design of high-performance catalytic systems.

5. Conclusions

In this work, high-performance $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$ ($0 < x < 1$) microspheres were developed for the FTO reaction and its dynamic structural evolution was revealed by *in/ex situ* characterizations. The volcano-shaped activity reaches a summit over Fe-13Mn and arises from the combined effect of the grain sizes and the content of active Fe carbides. Aside from the superior activity, Fe-13Mn exhibited the highest selectivity of 54.4% and STY of $73.0 \text{ g} \cdot \text{kg}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$ to lower olefins. Mn could substitutionally dissolve with Fe in the solid solution, expand the lattice spacing and reduce the surface energy of ferrite, which jointly leads to the lattice distortion and the poor crystallinity of $\text{Mn}_x\text{Fe}_{3-x}\text{O}_4$. Additionally, the expansion of the lattice spacing could decrease the Gibbs energy of Mn (II)-wüstite formation and stabilize the interstitial dissolved carbon atoms, thus stabilizing $\text{Mn}_x\text{Fe}_{1-x}\text{O}$ and $\varepsilon\text{-Fe}_2\text{C}$ during the FTO reaction. Furthermore, Mn is prone to enrich on the Fe surface and can strengthen CO dissociation by endowing Fe with strong surface basicity. Concurrently, Mn could weaken the hydrogenation of carbonaceous intermediates and hinder the readsorption and secondary reactions of olefins, thus steering the selectivity toward lower olefins.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest associated with the work submitted.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcat.2019.10.034>.

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